

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

**DEPARTMENT OF MECHANICAL AND PRODUCTION
ENGINEERING**

Assignment-1

**COURSE NO: ME4205
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Objective

The objective of this project is to develop a universal LabVIEW program capable of performing heat, vibration, and flow experiments. The program is designed to accept user inputs, process experimental data, and provide calculated outputs along with proper indications and visualizations. The goal is to integrate multiple experimental analyses into a single user-friendly interface.

Introduction

LabVIEW is a graphical programming platform widely used in engineering applications for data acquisition, analysis, and control. It allows users to design systems using a visual interface, making it highly suitable for experimental and automation-based tasks.

In this project, a unified LabVIEW program is developed to perform three distinct experiments: heat transfer, vibration analysis, and fluid flow analysis.

The heat experiment is based on the fundamental relation:

$$q = mc\Delta T$$

where heat transfer depends on mass, specific heat, and temperature difference.

The vibration experiment focuses on combining two signals with different frequencies and amplitudes. The resulting signal is analyzed using frequency-domain techniques such as Fast Fourier Transform (FFT) and power spectral density.

The flow experiment involves calculating flow rate, velocity, and Reynolds number using:

$$Q = \frac{\pi d^4}{128\mu L} \Delta P$$

$$V = \frac{4Q}{\pi d^2}$$

$$Re = \frac{\rho V d}{\mu}$$

Based on Reynolds number, the flow is classified as laminar, turbulent, or transitional.

This integrated approach enhances efficiency and provides a comprehensive tool for analyzing multiple engineering phenomena within a single platform.

Heat Experiment

The user can choose the mass(kg) value and final temperature(C) in the front panel as seen on figure 5 and the output is shown as the heat amount and a light indicating whether heat is being absorbed or released.

Final Temperature > Initial Temperature

Condition 1 :Here,final temperature is bigger than initial temperature.

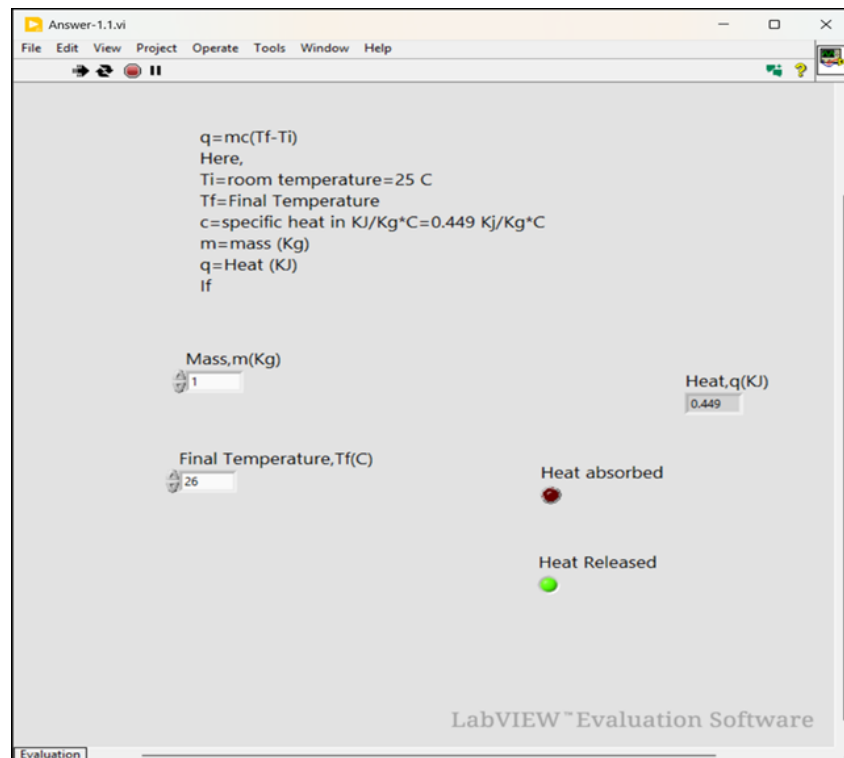


Figure 1: Front panel for heat experiment (heat released condition)

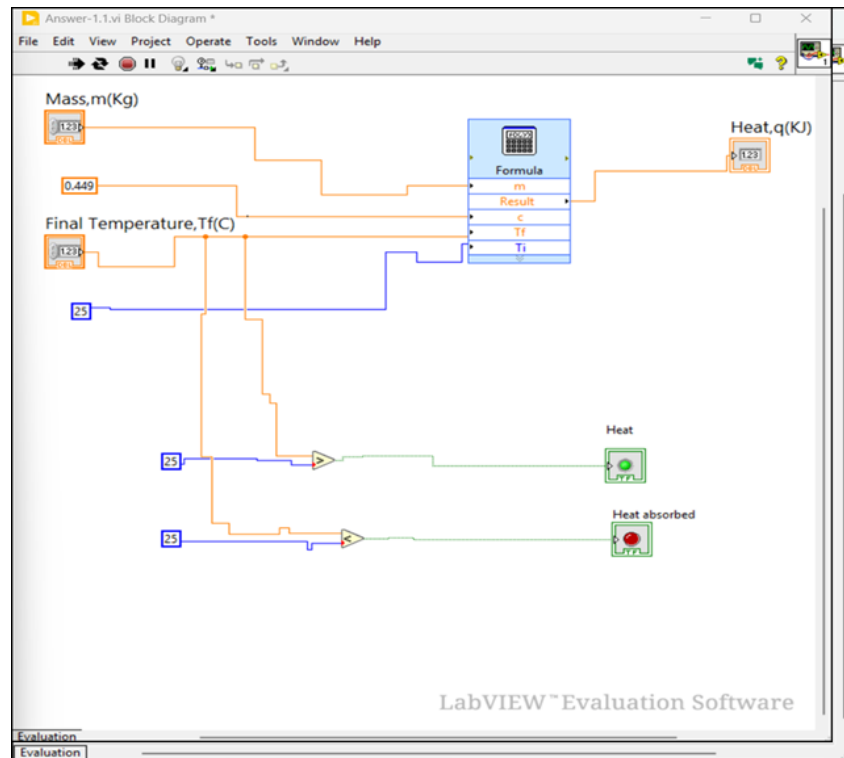


Figure 2: Block diagram for heat experiment

Final Temperature < Initial Temperature

Condition 2 :Here,final temperature is smaller than initial temperature.

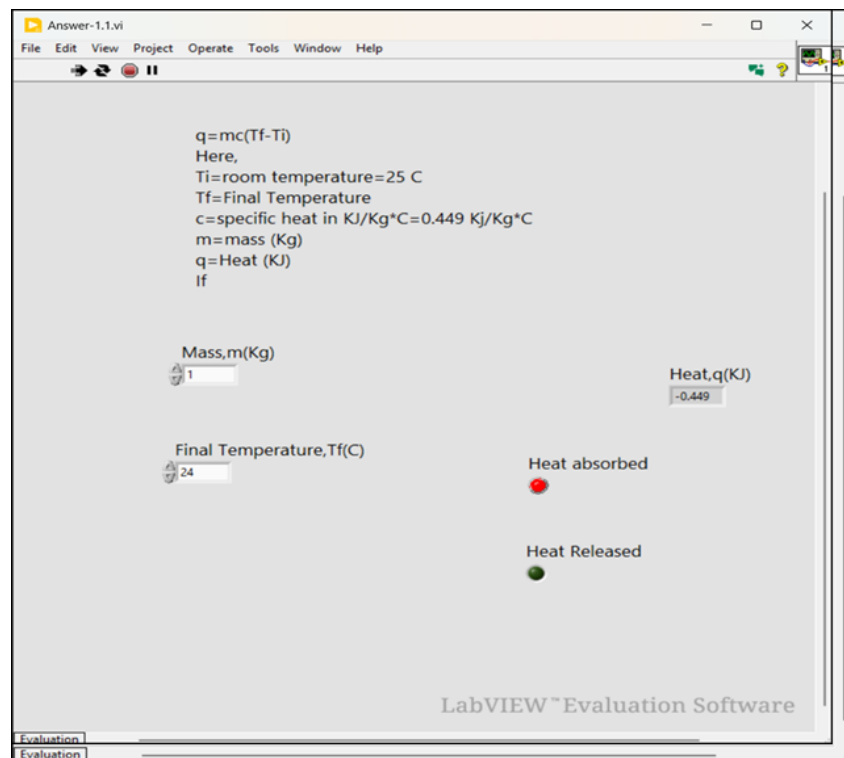


Figure 3: Front panel for heat experiment (heat absorbed condition)

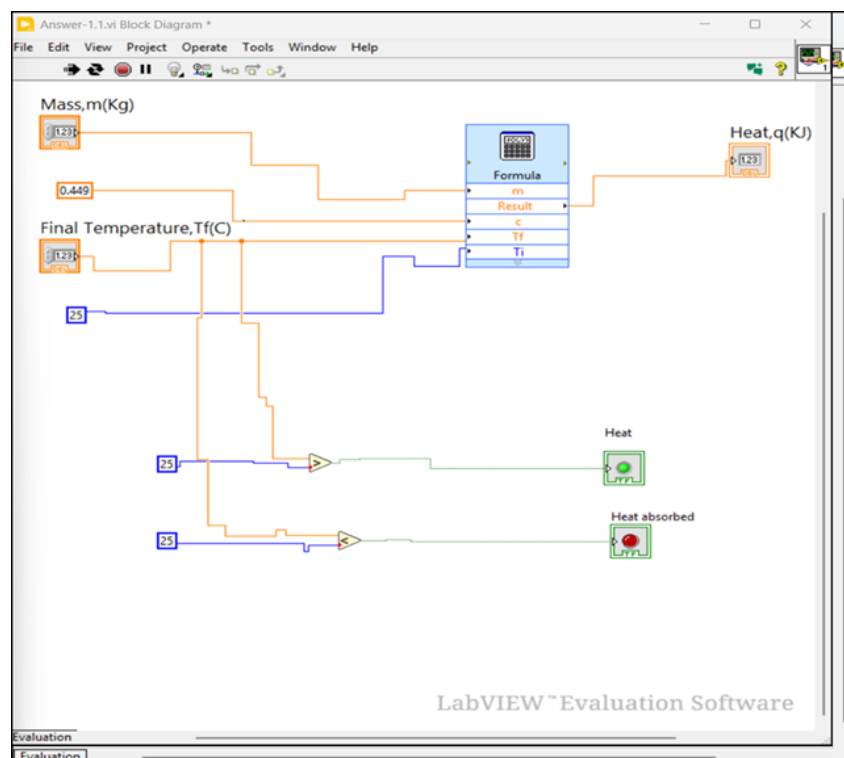


Figure 4: Block diagram for heat experiment

Final Temperature = Initial Temperature

Condition 3 :Here final temperature is equal to initial temperature.

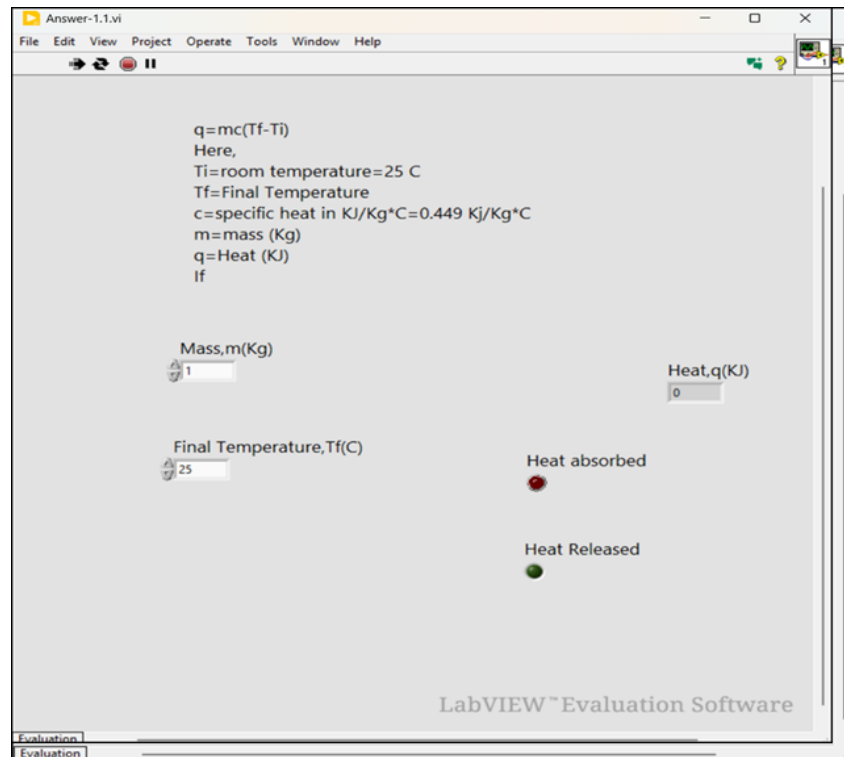


Figure 5: Front panel for equilibrium condition

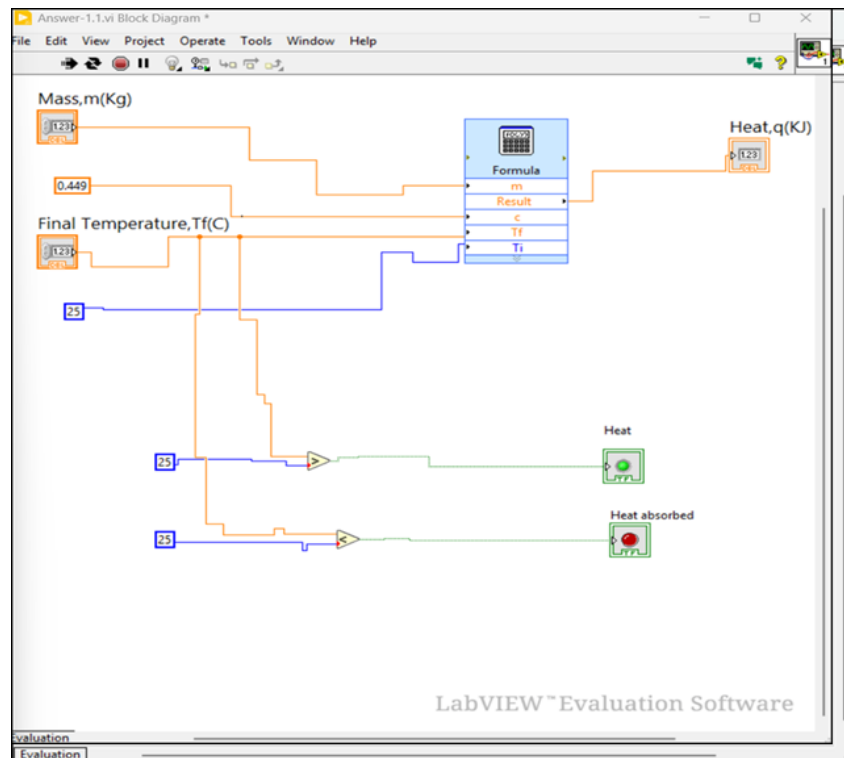


Figure 6: Block diagram for equilibrium condition

Vibration Experiment

User can choose to either add or multiply the two given signals, the numerical buttons are there to input both signals amplitude and frequency, Time graph visualizes the adding or multiplying operation of two signals, FFT Peak and Power Spectral Density waveform graph shows fft peak and power spectral density respectively. A save data button is given to save the data of three generated plots if the user wants to. And finally a stop button is to stop the program/loop completely.

Adding Operation

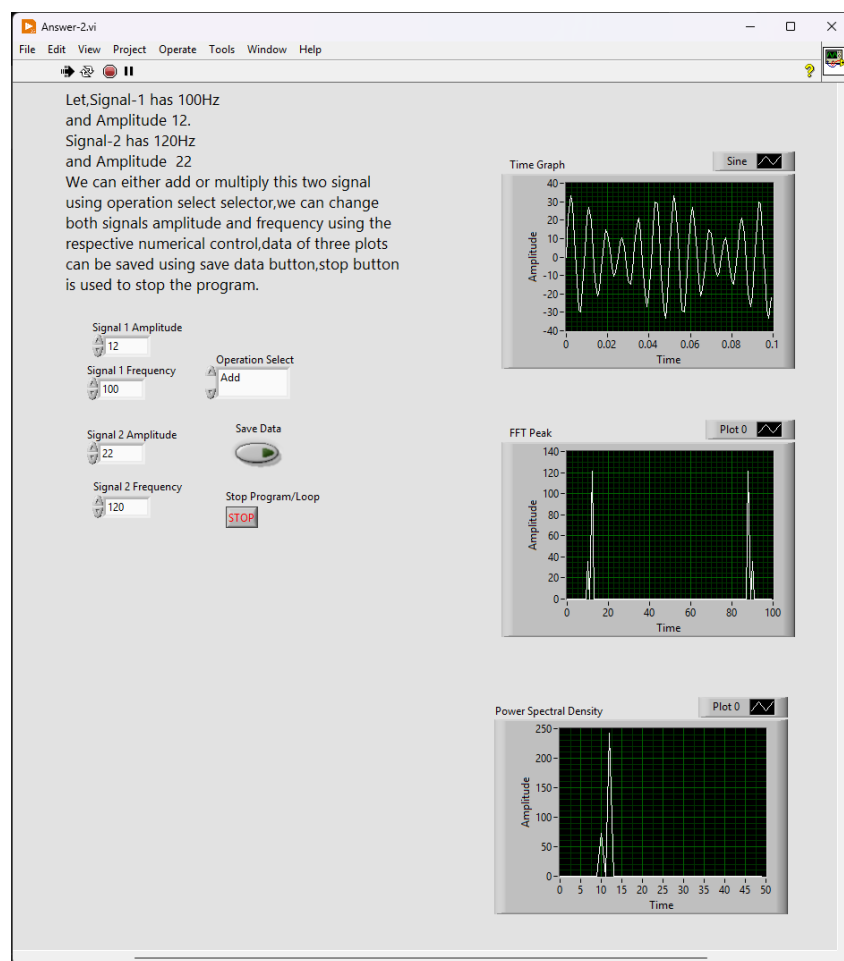


Figure 7: Front panel for addition of signals

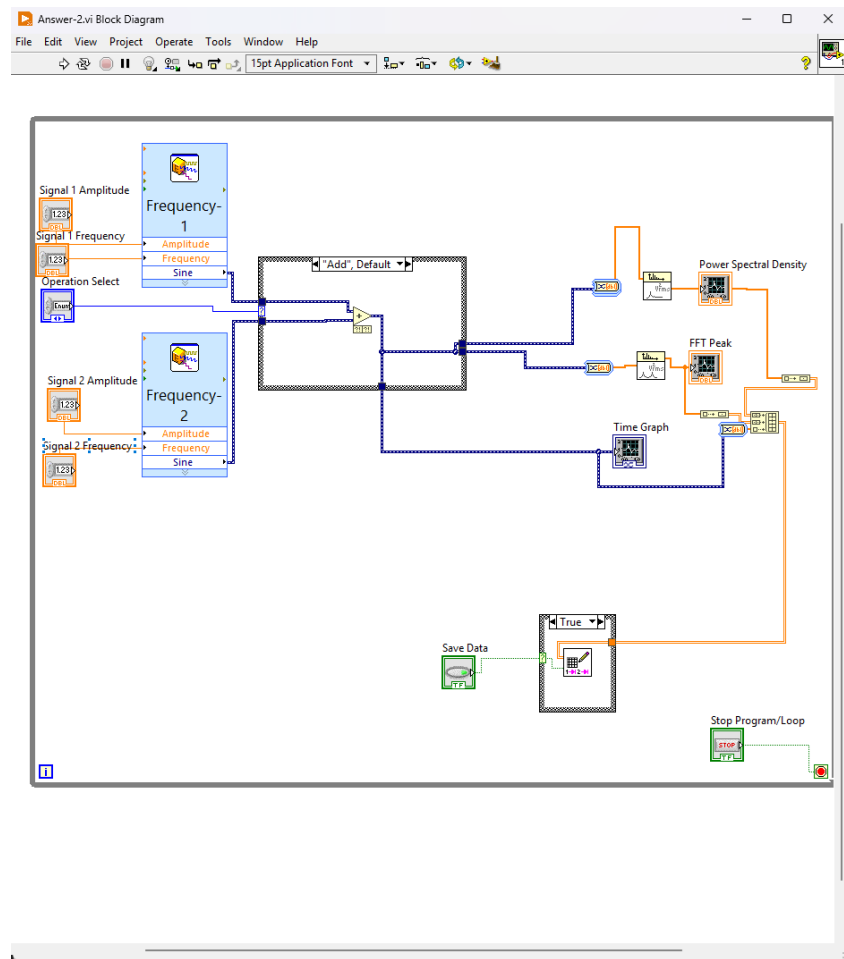


Figure 8: Block diagram for addition operation

Multiplying Operation

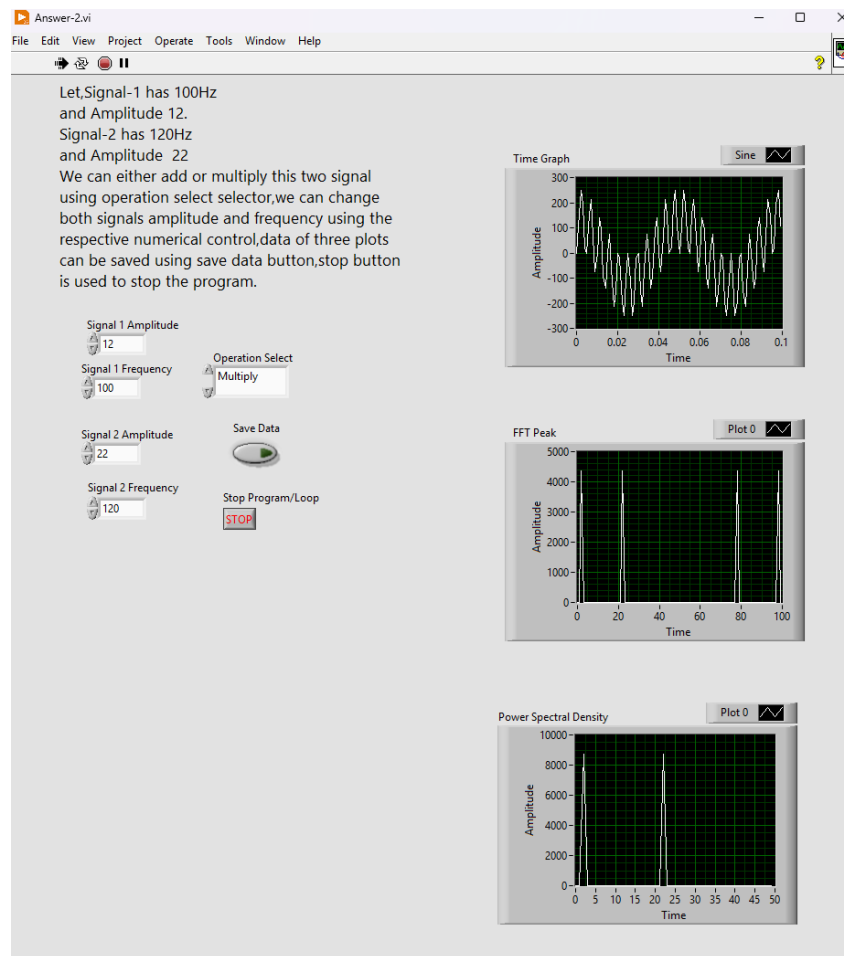


Figure 9: Front panel for multiplication of signals

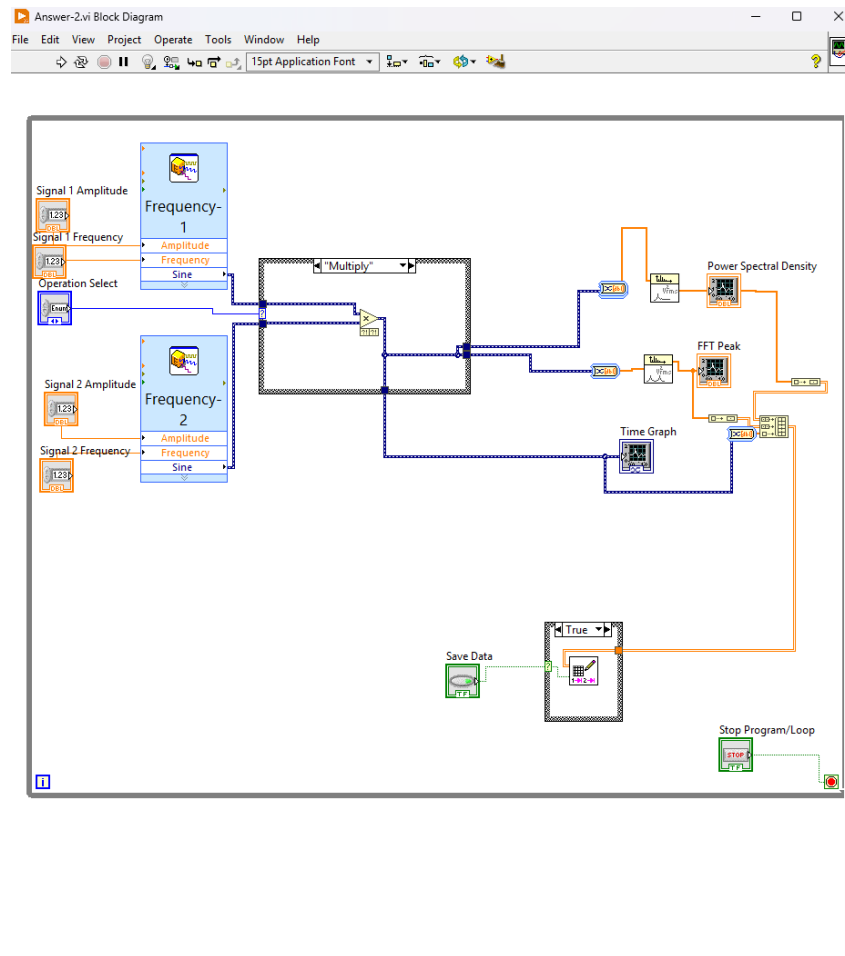


Figure 10: Block diagram for multiplication operation

Flow Experiment

In this problem, user can enter the length of the pipe, pressure difference, diameter of the pipe and the program will calculate the flowrate Q , Velocity V and Reynolds Number, R as well as will indicate with respective light what kind of flow it is. Finally if the Reynolds number is more than 25000 a warning light will be lit to indicate calculated flow rate maybe incorrect which can be seen on figure 17.

Turbulent Flow

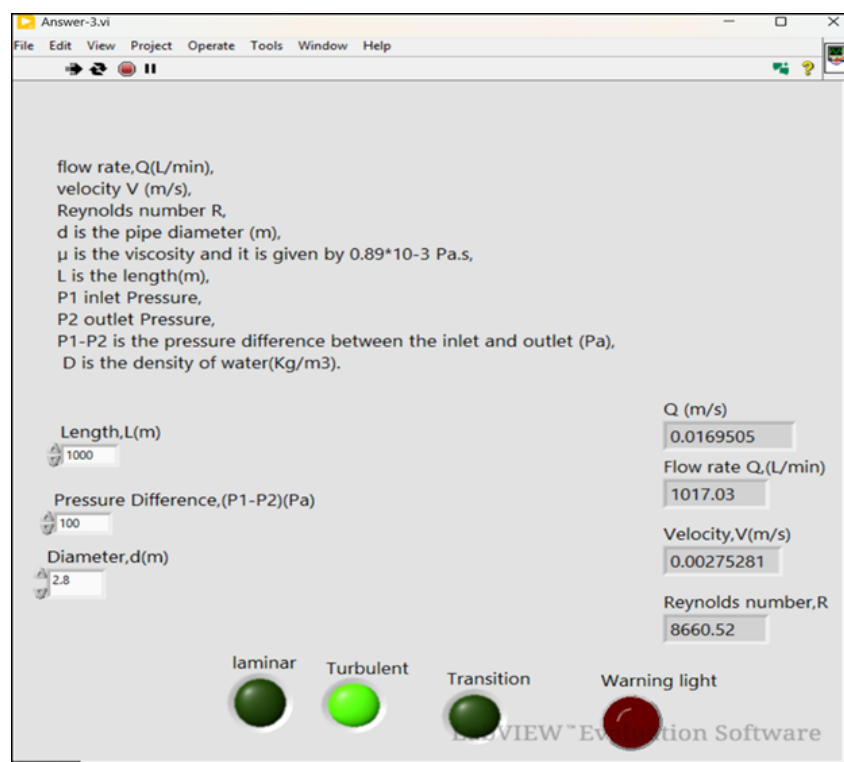


Figure 11: Front panel for turbulent flow

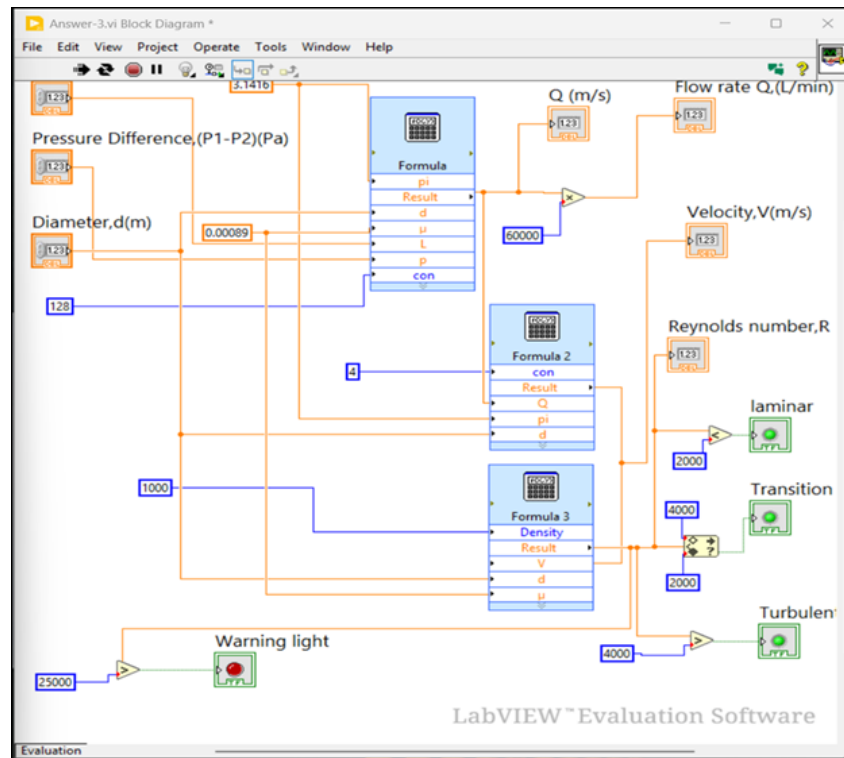


Figure 12: Block diagram for turbulent flow

Laminar Flow

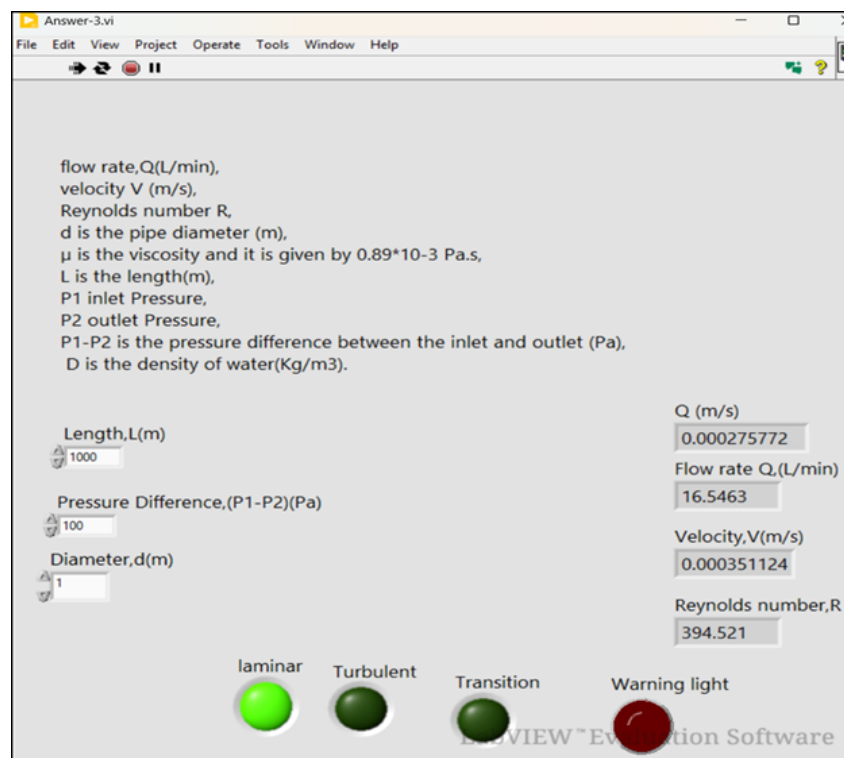


Figure 13: Front panel for laminar flow

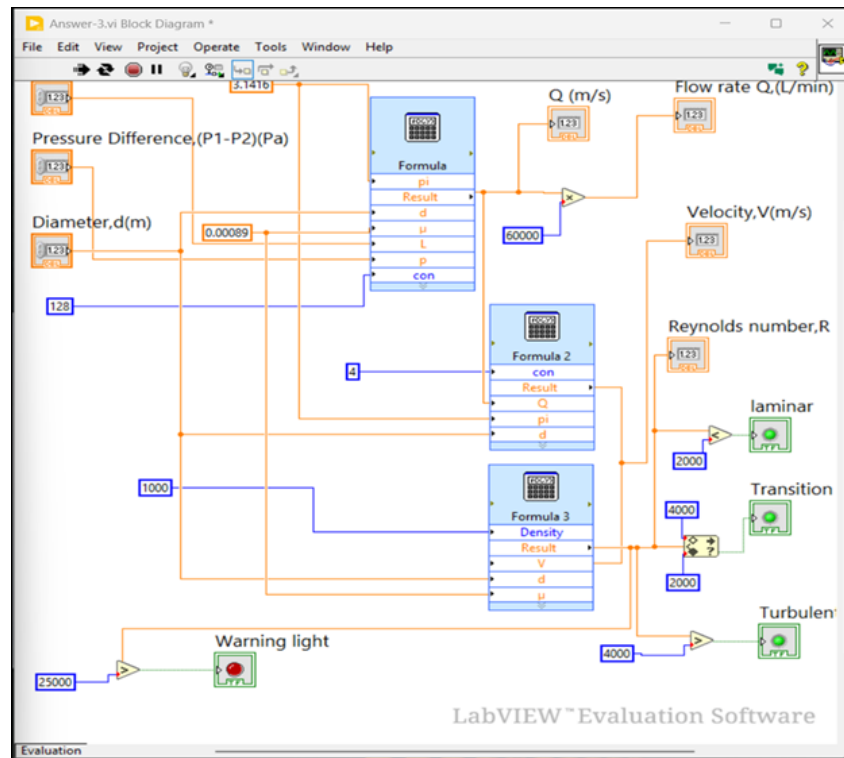


Figure 14: Block diagram for laminar flow

Transitional Flow

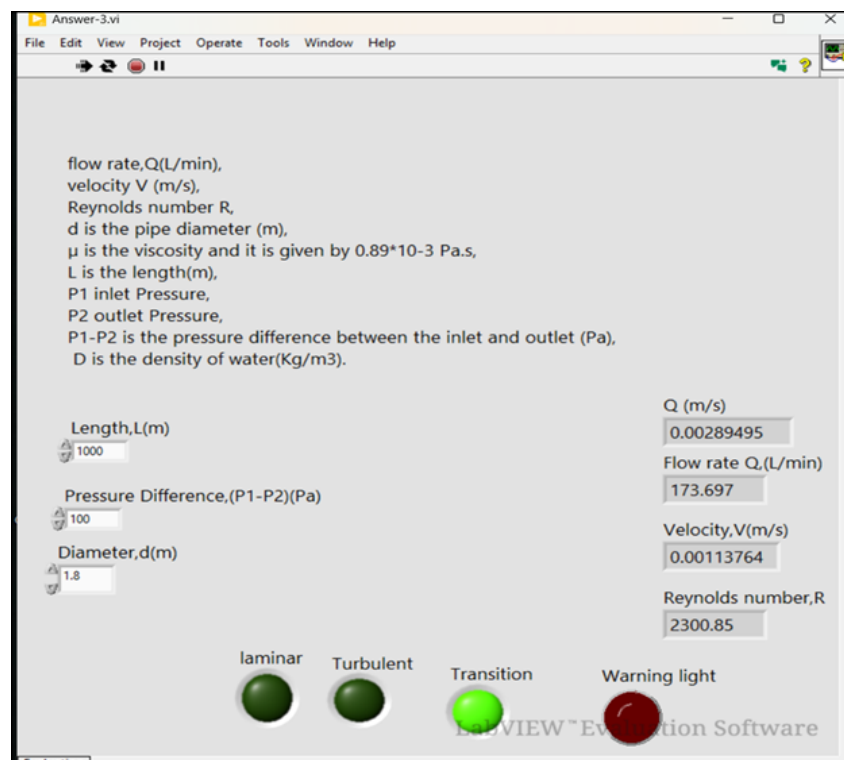


Figure 15: Front panel for transitional flow

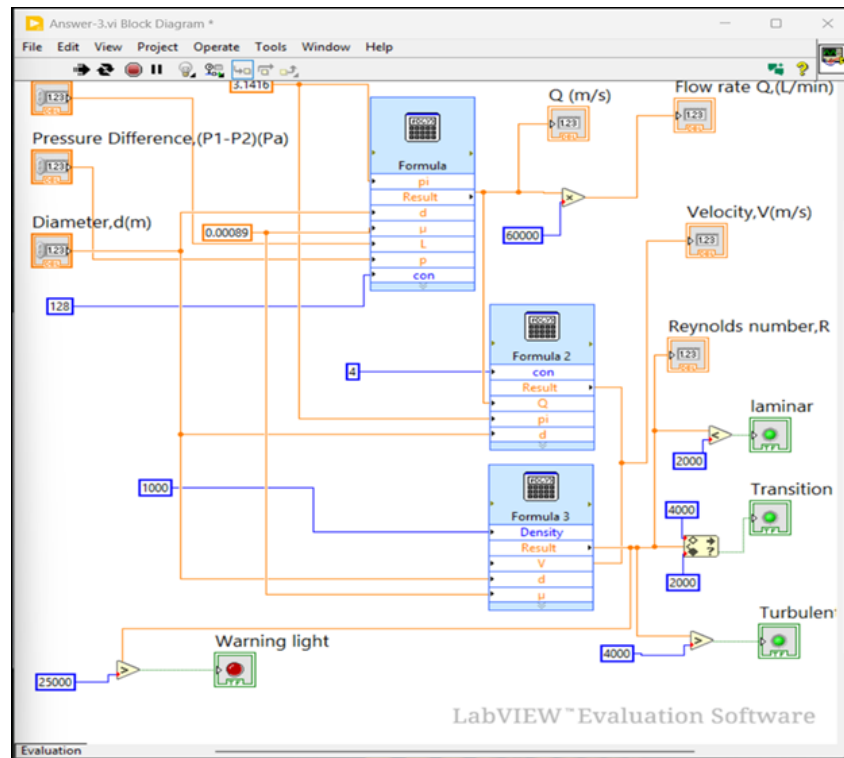


Figure 16: Block diagram for transitional flow

Warning Condition

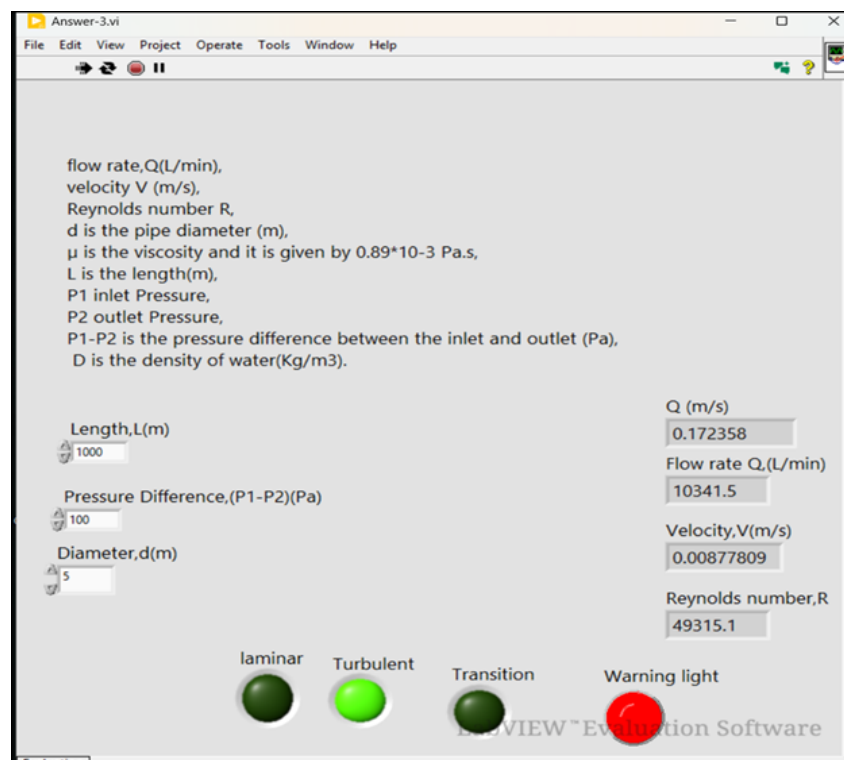


Figure 17: Front panel showing warning condition

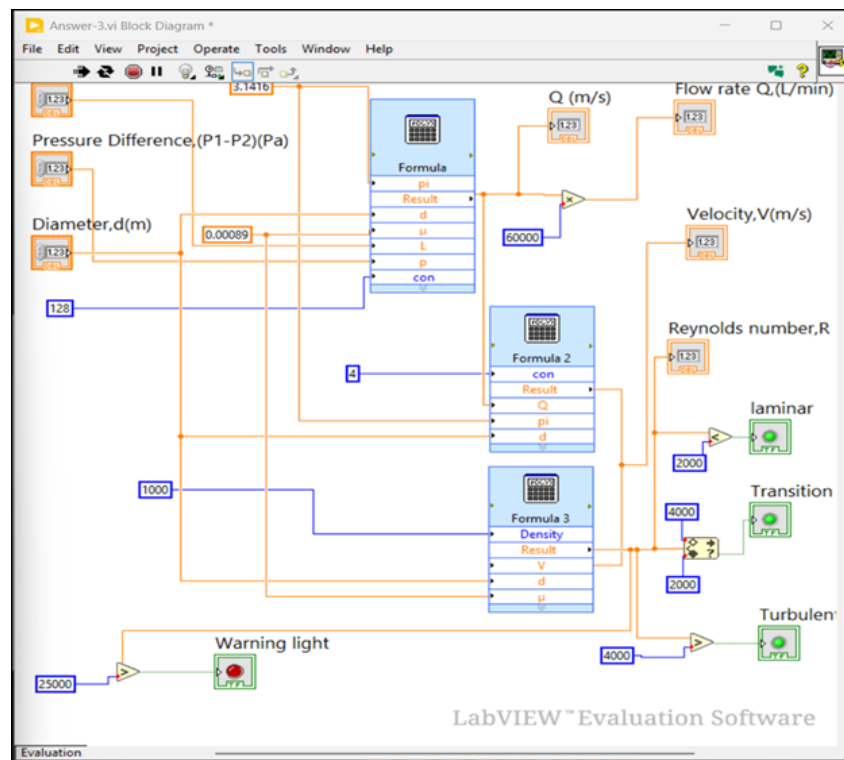


Figure 18: Block diagram showing warning condition

Conclusion

In this project, a comprehensive LabVIEW program was successfully developed to perform heat, vibration, and flow experiments within a single interface. The program effectively processes user inputs, performs necessary calculations, and provides accurate outputs along with visual indicators and graphical representations.

All required conditions, including heat absorption/release, signal operations, and flow classification, were properly implemented. The system demonstrates both functional accuracy and user-friendly design, fulfilling all assignment requirements.

References

1. LabVIEW User Manual, National Instruments.
2. Course Materials for ME4205 (Mechatronics).